

MINSK MAZOWIECKI, Poland

WMO: 123760

Lat: 52.1922N

Lon: 21.6635E

Elev: 604

StdP: 14.38

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	7.3	-4.1	4.5	3.1	1.7	6.1	9.0	27.1	39.0	23.9	37.2	5.2	90	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	86.7	68.7	83.3	66.7	80.2	65.4	70.7	82.2	69.1	79.8	67.3	76.8	6.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	101.7	75.5	65.5	96.2	73.4	63.9	90.9	71.9	34.9	82.6	33.6	80.2	32.1	76.7	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.5	18.3	15.9	-5.4	92.6	8.2	2.4	-11.2	94.3	-16.0	95.7	-20.6	97.1	-26.5	98.8		
			-4.1	73.8	8.3	2.0	-10.1	75.3	-15.0	76.4	-19.7	77.6	-25.8	79.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	26.3	28.7	36.7	47.8	56.6	62.6	67.3	65.8	56.9	47.0	39.3	31.7
	DBStd	16.04	10.75	9.72	9.02	7.70	7.31	6.08	5.98	5.92	6.30	7.33	7.01	8.85
	HDD50	2947	735	598	416	128	21	1	0	0	11	148	323	566
	HDD65	6725	1200	1018	876	516	277	116	46	61	255	559	770	1031
	CDD50	1973	0	0	5	63	226	379	535	491	217	54	3	0
	CDD65	275	0	0	0	1	16	44	116	87	11	0	0	0
	CDH74	2636	0	0	0	27	192	454	1042	814	107	0	0	0
	CDH80	732	0	0	0	5	35	103	319	251	19	0	0	0
Wind	WSAvg	7.2	8.0	7.8	8.3	7.8	6.9	6.7	6.5	5.9	6.5	7.1	7.5	8.0
Precipitation	PrecAvg	22.9	1.3	1.2	1.3	1.5	2.3	2.7	3.2	2.6	2.1	1.5	1.6	1.5
	PrecMax	30.4	3.8	2.7	2.8	3.9	5.3	5.9	11.0	8.3	5.6	4.7	4.3	3.0
	PrecMin	17.3	0.2	0.2	0.4	0.1	0.9	0.7	0.8	0.3	0.5	0.2	0.1	0.4
	PrecStd	2.8	0.7	0.5	0.6	0.8	1.0	1.3	2.0	1.5	1.4	1.0	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	50.9	63.7	77.5	84.1	87.8	90.8	91.9	82.9	70.2	58.9	50.9
		MCWB	45.4	46.1	51.7	59.2	64.7	70.3	69.0	70.0	65.0	59.7	52.9	47.0
	2%	DB	44.9	47.2	57.8	71.2	79.2	82.7	86.9	86.2	77.0	65.6	54.2	47.4
		MCWB	42.2	43.0	47.6	55.8	63.4	67.1	68.2	69.2	62.3	57.3	50.0	45.2
	5%	DB	42.3	43.7	53.7	66.5	74.8	79.2	83.6	82.2	72.7	61.6	50.9	44.7
		MCWB	40.1	40.3	45.6	53.2	60.9	64.6	67.0	66.6	60.7	55.2	48.4	42.8
	10%	DB	39.5	40.4	49.5	62.2	70.2	75.6	80.1	78.5	68.7	58.0	48.7	42.0
		MCWB	37.7	37.9	43.6	50.5	58.1	62.7	65.9	64.9	58.8	53.5	46.6	40.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.8	47.2	52.7	60.0	67.5	72.1	73.0	72.9	67.0	61.2	53.8	48.1
		MCDB	47.4	49.9	60.6	74.5	79.5	84.3	85.0	85.3	78.9	68.4	57.9	49.8
	2%	WB	42.7	43.4	49.2	57.2	64.6	69.3	70.9	70.3	64.1	58.3	51.0	45.5
		MCDB	44.7	46.3	55.9	68.2	75.6	79.7	82.1	82.2	74.0	63.5	53.4	47.0
	5%	WB	40.3	40.5	46.7	54.3	62.3	66.9	69.5	68.2	61.9	56.0	48.8	42.9
		MCDB	42.0	43.5	52.0	64.3	72.6	76.1	79.9	79.2	69.8	60.4	50.6	44.4
	10%	WB	37.6	38.1	43.9	51.6	60.1	64.4	67.8	66.5	60.0	53.9	46.8	40.6
		MCDB	39.4	40.3	48.8	59.8	68.5	73.0	76.8	75.8	66.9	57.7	48.4	41.9
Mean Daily Temperature Range	5% DB	MDBR	8.6	10.4	14.3	19.1	19.5	20.1	19.8	20.0	17.7	14.3	8.7	8.0
		MCDBR	8.8	13.1	21.1	26.0	25.1	24.9	24.8	25.6	24.1	19.6	12.8	9.8
		MCWBR	7.6	9.7	12.7	13.4	12.1	11.9	9.9	11.0	11.6	11.9	9.5	8.1
	5% WB	MCDBR	8.5	12.2	19.2	24.5	23.0	22.2	21.6	22.5	21.1	17.6	11.5	9.3
		MCWBR	7.6	9.5	12.5	13.2	11.9	11.3	9.9	10.3	11.0	11.3	9.1	8.1
Clear-Sky Solar Irradiance	taub		0.314	0.342	0.375	0.428	0.412	0.409	0.439	0.433	0.394	0.365	0.354	0.312
	taud		2.225	2.215	2.219	2.134	2.241	2.280	2.212	2.223	2.302	2.358	2.282	2.269
	Ebn at Noon		210	239	255	255	266	268	257	251	247	230	193	190
	Edh at Noon		22	29	36	44	41	40	42	40	33	25	21	18
All-Sky Solar Radiation	RadAvg		221	451	837	1295	1645	1744	1629	1427	1008	562	235	157
	RadStd		43	74	114	174	166	150	193	117	153	91	23	23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	+0.90	N/A	N/A	N/A	+0.74	N/A	N/A	-297	+144	+50

Nomenclature: See separate page